Assignment Number One

First Unit: Number Systems, Bitwise Operations, Integer Representations & Truth Tables (30 Questions)

Part 1: Fundamentals & Formulas Summary

- **Number Systems & Arithmetic:** Conversions via successive division/positional sums ($\sum bit \times 2^i$). Basic operations in binary & decimal.
- **Bitwise Operators & Truth Tables:** AND (&), OR (|), XOR (^), NOT (~), NAND, NOR. Truth tables map every combination of binary inputs to outputs.
- **Optimization Tricks:** Even/odd checking ($N \ \& \ 1$), XOR swap without temporary variables, Brian Kernighan's bit-counting algorithm.
- **Integer Representations:** Unsigned, Sign-Magnitude, 1's, and 2's complement ranges, overflow, underflow, and wrap-around behavior.

Part 2: Assignment Question Paper (30 Questions)

Q1: Convert decimal 215 to binary.

Q2: Convert binary 11011011_2 to decimal.

Q3: Perform binary addition: $10111_2 + 1101_2$.

Q4: Perform binary subtraction: $11010_2 - 1011_2$.

Q5: Perform binary multiplication: $1011_2 \times 101_2$.

Q6: Given $A = 45$ and $B = 25$, compute $A \ \& \ B$ and $A \mid B$.

Q7: Compute bitwise XOR $A \wedge B$ and NOT $\sim A$ (8-bit) for Q6.

Q8: Evaluate NAND and NOR for $X = 12$ and $Y = 10$.

Q9: Calculate $19 << 3$ and $76 >> 2$ with equivalents.

Q10: Write bitwise expression to check even/odd; test for $N = 37$.

Q11: Demonstrate XOR swap algorithm for $x = 14$, $y = 22$.

Q12: Trace Brian Kernighan's algorithm for $N = 43$.

Q13: State 16-bit ranges for unsigned, sign-magnitude, 2's complement.

Q14: Represent -68 in 8-bit Sign-Magnitude.

Q15: Represent -68 in 8-bit 1's complement.

Q16: Represent -68 in 8-bit 2's complement.

Q17: Convert 8-bit 2's complement 10110101_2 to decimal.

Q18: Compare Sign-Magnitude, 1's, and 2's complement.

Q19: Explain signed overflow: $+75 + +60$ in 2's complement.

Q20: Describe underflow/wrap-around for $3 - 10$ (8-bit unsigned).

Q21: Construct the 2-input truth table for Bitwise AND ($A \ \& \ B$).

Q22: Construct the 2-input truth table for Bitwise OR ($A \mid B$).

Q23: Construct the 2-input truth table for Bitwise XOR ($A \wedge B$).

Q24: Construct the 1-input truth table for Bitwise NOT ($\sim A$).

Q25: Construct the 2-input truth table for NAND operation.

Q26: Construct the 2-input truth table for NOR operation.

Q27: Construct a 3-variable truth table for $F = (A \ \& \ B) \mid (\sim C)$.

Q28: Construct the truth table for Half-Adder Sum ($A \oplus B$).

Q29: Construct the truth table for Half-Adder Carry ($A \land B$).

Q30: Verify De Morgan's Law via truth table: $\sim(A \mid B) \equiv \sim A \ \& \ \sim B$.

Part 3: Step-by-Step Solutions (Q1 to Q30)

Solutions 1 to 5: Number Systems & Arithmetic

Sol 1: Successive division of 215 by 2 yields remainders: 11010111_2.

Sol 2: Positional expansion of 11010111_2: $128 + 64 + 16 + 4 + 2 + 1 = 215$.

Sol 3: Binary addition $10111_2\ (23) + 1101_2\ (13) = 100100_2$ (Decimal 36).

Sol 4: Binary subtraction $11010_2\ (26) - 1011_2\ (11) = 01111_2$ (Decimal 15).

Sol 5: Binary multiplication $1011_2 \times 101_2 = 110111_2$ (Decimal 55).

Solutions 6 to 10: Bitwise & Shift Operations

Sol 6: $A = 00101101$, $B = 00011001$. $A \& B = 00001001_2$ (9); $A \mid B = 00111101_2$ (61).

Sol 7: $A \wedge B = 00110100_2$ (52); $\sim A = 11010010_2$ (-46).

Sol 8: $X = 1100$, $Y = 1010$. $NAND = \sim(1000) = 0111_2$ (7); $NOR = \sim(1110) = 0001_2$ (1).

Sol 9: $19 << 3 = 19 \times 8 = 152$; $76 >> 2 = 76 / 4 = 19$.

Sol 10: Expression: $(N \& 1) == 0$. For 37 (100101_2), $37 \& 1 = 1$ (Odd).

Solutions 11 to 15: Optimization, Ranges & Representations

Sol 11: XOR swap for $x = 14$, $y = 22$: (1) $x = x \wedge y\ (24)$; (2) $y = x \wedge y\ (14)$; (3) $x = x \wedge y\ (22)$.

Sol 12: Brian Kernighan's for $43\ (101011_2)$ clears LSB each step: 43&42, 42&41, 40&39, 32&31 $\rightarrow$ Total set bits = 4.

Sol 13: 16-bit ranges: Unsigned $[0, 65535]$; Sign-Magnitude $[-32767, 32767]$; 2's Complement $[-32768, 32767]$.

Sol 14: 8-bit Sign-Magnitude for -68: Magnitude $68 = 01000100_2$, set sign bit $\rightarrow$ 11000100_2.

Sol 15: 8-bit $1's$ Complement for -68: Invert 01000100_2 $\rightarrow$ 10111011_2.

Solutions 16 to 20: 2's Complement, Overflow & Underflow

Sol 16: 8-bit $2's$ Complement for -68: $10111011 + 1 = 10111100_2$.

Sol 17: 2's complement 10110101_2 to decimal: Invert and add 1 $\rightarrow$ magnitude $01001011_2\ (75)$ $\rightarrow$ -75.

Sol 18: Comparison: Sign-Magnitude and $1's$ complement have dual zeros ($+0/-0$) and complex hardware; $2's$ complement has a single zero and optimal arithmetic.

Sol 19: Signed overflow: $+75\ (01001011) + +60\ (00111100) = 10000111_2\ (-121)$, exceeding $+127$ limit.

Sol 20: Unsigned underflow: $3 - 10 = -7$ $\rightarrow$ wraps around to $256 - 7 = 249$ (11111001_2).

Solutions 21 to 30: Truth Tables & Logic Concepts

Q21 (AND):

A	B	A & B
0	0	0
0	1	0
1	0	0
1	1	1

Q22 (OR):

A	B	A | B
0	0	0
0	1	1
1	0	1
1	1	1

Q23 (XOR):

A	B	A ^ B
0	0	0
0	1	1
1	0	1
1	1	0

Q24 (NOT):

A	~A
0	1
1	0

Q25 (NAND):

A	B	NAND
0	0	1
0	1	1
1	0	1
1	1	0

Q26 (NOR):

A	B	NOR
0	0	1
0	1	0
1	0	0
1	1	0

Q27 ($F = (A\&B)|(\sim C)$):

A	B	C	F
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

Q28 & Q29 (Half-Adder Sum & Carry): Sum = $A \oplus B$, Carry = $A \land B$.

A	B	Sum (A ^ B)	Carry (A & B)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Q30 (De Morgan's Verification $\sim(A \mid B) \equiv \sim A \,\&\, \sim B$):

A	B	A | B	~(A | B)	~A	~B	~A & ~B
0	0	0	**1**	1	1	**1**
0	1	1	**0**	1	0	**0**
1	0	1	**0**	0	1	**0**
1	1	1	**0**	0	0	**0**

Since columns 4 and 7 are identical, De Morgan's Law is verified.